Option C – The Physics of Sports

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
9	(a)		Substitution into principle of moments i.e. $F\cos 8 \times 0.07 = (12.9 \times 0.19) + (74 \times 0.39)$ (1) F = 451 [N] (1) If $F\sin 7 \times 0.07$ **or** $F \times 0.07$ used correctly – award 1 mark	1	1		2	1	
	(b)		Recall $Ft = mv - mu$ or equivalent (1) Initial momentum = 907 N s and impulse = +/- 968 N s (1) Final velocity = - 0.52 [m s^{-1}] (need sign) (1) **Alternative:** Recall $F = ma$ (1) Acceleration = 47 m s^{-2} and use of $v = u + at$ (1) Final velocity = - 0.52 [m s^{-1}] (need sign) (1)	1	1 1		3	2	
	(c)	(i)	Recall drag force $F = \frac{1}{2}\rho v^2 A C_D$ (1) Density and drag coefficient are constant and increasing velocity increases the magnitude of F (1) accept density can be variable with altitude [Effective / cross-sectional / surface] area of A is less than B so drag [force] is reduced (1)	1	1 1		3		
		(ii)	Rotational kinetic energy $= \frac{1}{2}I\omega^2$ (1) ω is the number of revolutions $\times 2\pi$ (or ω is proportional to revs per second) so will not double (1)			2	2		